

FLISA II, Norway

WMO: 013920

Lat: 60.6139N

Lon: 12.0122E

Elev: 185

StdP: 99.12

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -21.7	(c) -19.2	(d) -24.2	(e) 0.4	(f) -21.7	(g) -21.6	(h) 0.6	(i) -19.2	(j) 6.3	(k) 1.3	(l) 5.8	(m) 0.9	(n) 0.9	(o) 30	(p) 0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.9	(c) 27.1	(d) 16.4	(e) 25.1	(f) 16.0	(g) 22.9	(h) 15.0	(i) 17.9	(j) 24.1	(k) 17.0	(l) 22.4	(m) 16.2	(n) 21.0	(o) 2.2	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.9	11.5	19.4	15.0	10.9	18.7	14.1	10.3	17.9	51.1	24.1	48.4	22.5	45.9	21.2	21.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.6	4.9	4.3	-24.1	29.1	3.4	2.0	-26.5	30.6	-28.5	31.7	-30.4	32.8	-32.8	34.2
			-24.5	19.4	3.4	1.2	-27.0	20.2	-28.9	20.9	-30.8	21.6	-33.2	22.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	-5.9	-4.9	-0.6	5.1	10.3	14.3	17.0	15.0	11.0	4.9	0.1	-4.7
	DBStd	9.12	6.03	5.61	4.86	3.40	4.12	3.11	2.96	2.43	3.09	3.83	4.54	6.94
	HDD10.0	2379	492	417	328	153	47	2	0	1	24	162	298	455
	HDD18.3	4827	750	650	587	398	252	127	64	105	219	415	548	713
	CDD10.0	626	0	0	0	5	56	132	217	157	56	5	0	0
	CDD18.3	33	0	0	0	0	2	6	22	3	0	0	0	0
	CDH23.3	378	0	0	0	0	33	91	231	23	0	0	0	0
	CDH26.7	67	0	0	0	0	2	15	50	1	0	0	0	0
Wind	WSAvg	1.9	1.7	1.7	1.9	2.0	2.1	2.1	1.9	1.8	1.8	1.8	1.8	1.7
Precipitation	PrecAvg	701	46	33	33	42	61	77	81	90	68	68	59	50
	PrecMax	881	101	69	89	101	166	137	162	162	143	149	108	90
	PrecMin	556	5	10	2	4	9	4	16	36	19	12	19	5
	PrecStd	92	26	16	18	22	37	34	37	30	27	31	25	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	6.8	13.9	19.7	26.6	28.6	30.3	25.8	21.7	15.7	10.6	8.6
		MCWB	4.5	3.4	7.3	10.4	14.8	16.1	18.0	16.6	14.5	11.8	8.9	6.8
	2%	DB	4.2	4.7	10.4	17.3	23.8	26.1	28.1	23.6	19.5	13.6	9.1	6.8
		MCWB	2.9	2.7	5.3	9.6	13.7	15.3	17.4	15.8	13.6	10.5	7.5	5.4
	5%	DB	3.0	3.3	7.7	14.7	21.2	23.2	26.1	22.0	17.9	12.1	7.6	5.3
		MCWB	1.9	1.9	3.5	8.1	12.8	14.5	17.0	15.2	12.9	9.6	6.3	3.9
	10%	DB	1.7	2.1	5.8	12.1	18.2	21.0	23.8	20.2	16.2	10.7	6.2	3.8
		MCWB	0.6	0.9	2.6	6.5	11.8	13.3	16.0	14.5	12.4	8.7	5.2	2.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	4.5	7.7	11.5	16.5	17.6	19.6	18.2	16.3	13.1	9.3	7.2
		MCDB	5.6	5.9	12.9	17.4	22.4	23.4	26.1	22.0	18.9	14.7	10.3	8.3
	2%	WB	3.2	3.1	5.6	10.1	15.0	16.4	18.5	17.2	14.9	11.3	7.7	5.6
		MCDB	4.0	4.2	9.6	16.1	21.5	23.2	25.4	21.2	17.8	13.0	8.8	6.7
	5%	WB	1.9	2.0	4.2	8.7	13.6	15.3	17.6	16.3	13.9	10.1	6.5	4.1
		MCDB	2.8	3.0	7.0	14.2	19.4	21.8	24.3	20.0	16.8	11.6	7.5	5.1
	10%	WB	0.8	1.0	3.0	7.1	12.1	14.2	16.7	15.5	12.9	9.0	5.2	2.6
		MCDB	1.6	2.0	5.4	11.3	17.4	19.9	22.4	18.9	15.6	10.5	6.1	3.5
Mean Daily Temperature Range	5% DB	MDBR	5.5	6.5	9.2	10.4	10.3	10.0	9.9	9.1	8.9	6.6	5.0	5.6
		MCDBR	5.5	6.8	11.7	14.9	14.9	14.1	13.8	12.0	11.1	7.8	6.0	6.1
		MCWBR	4.5	5.2	7.2	8.1	6.8	6.2	5.6	5.6	6.3	5.4	5.3	5.5
	5% WB	MCDBR	4.8	6.0	10.7	13.5	12.3	12.4	12.1	9.6	9.0	6.3	5.6	5.9
		MCWBR	4.4	4.7	7.0	7.6	6.0	5.9	5.2	5.1	5.6	5.1	5.2	5.5
Clear-Sky Solar Irradiance	taub	0.242	0.287	0.314	0.354	0.356	0.348	0.368	0.359	0.327	0.313	0.262	N/A	
	taud	2.190	2.279	2.317	2.347	2.425	2.470	2.430	2.450	2.527	2.479	2.278	N/A	
	Ebn at Noon	529	711	813	842	870	883	852	831	803	686	501	N/A	
	Edh at Noon	38	64	86	102	102	99	101	91	71	54	35	N/A	
All-Sky Solar Radiation	RadAvg	0.28	0.92	2.40	3.83	4.73	5.38	4.82	3.77	2.49	1.11	0.35	0.17	
	RadStd	0.04	0.17	0.26	0.47	0.47	0.40	0.66	0.36	0.30	0.22	0.10	0.02	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page